

Monthly Intelligence Report

Edition 003 — The Osaka–Aichi Corridor

Where the energy transition is outrunning the grid. edition · June 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

5,981

299 EHV · 5,682 HV

GRID LINES MAPPED

~11,600

~105,000 km coverage

MC SIMULATIONS

78.9 M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero
proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 78.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 469,404 output data points with full uncertainty quantification across 5,981 substations and ~11,600 grid lines spanning ~105,000 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering Japan's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

亀田変電所

Kagawa, Shikoku · 66 kV

0.512

High

0.104

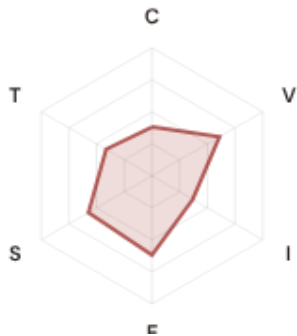
22th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE



COMPONENTS

- C** Continuity: 0.382 / 0.30 (127%)
- I** Infrastructure: 0.368 / 0.25 (147%)
- S** Saturation: 0.573 / 0.20 (287%)
- V** Voltage: 0.610 / 0.10 (610%)
- E** Economic: 0.618 / 0.10 (618%)
- T** Transition: 0.414 / 0.05 (828%)

MODIFIERS

- R3 Consequence:** 1.059 ↑ amplifies
- R4 Graph Criticality:** 0.994 → neutral
- R6a Restoration:** 1.000 → neutral
- R6b Seismic:** 1.057 ↑ amplifies
- R7 Digital Readiness:** 0.998 → neutral

This month's spotlight — automatically selected for June 2026 — is 亀田変電所 in Kagawa, Shikoku. With an R_final of 0.512 (High band, 22th fleet percentile), its risk profile is dominated by the T (Transition) component at 828% of its weight ceiling, followed by E (Economic) at 618%. Modifiers collectively shift R_base by 10.3%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Society 5.0 / Japan 2030 digitalisation planning requires.

CYBER HIGH EXPOSURE

0
0.0% of fleet

BLIND SPOTS

2236
High/Critical + R7 < median

AVG R7 MODIFIER

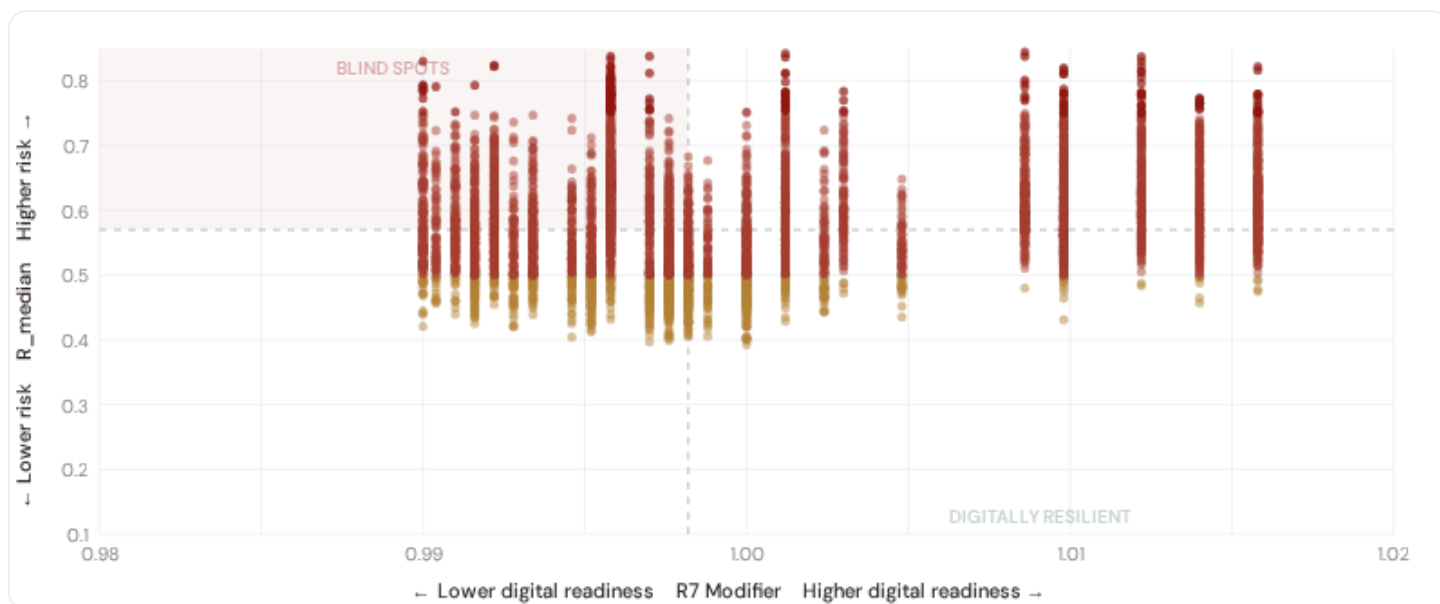
1.0012
Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

1,491
24.9% · R3 ≥ 1.15

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **2236 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 0.992$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Tohoku (686)**, **Hokkaido (403)**, **Kyushu (335)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

Capital-Intensive / High-Consequence

Est. VoLL: ¥2,000–4,000/kWh (JPY)

Capital nodes — highest economic loss per interruption (semiconductor fabs, data centers, refineries, hospitals)

1,491 substations (24.9%) High/Critical: **1467** (98.4%) Avg R: **0.627** Avg E: **0.585** / 0.10



Industrial / Medium-Large Enterprise

Est. VoLL: ¥1,000–2,000/kWh (JPY)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

1,513 substations (25.3%) High/Critical: **1423** (94.1%) Avg R: **0.603** Avg E: **0.586** / 0.10



Commercial / SME-Dense

Est. VoLL: ¥400–1,000/kWh (JPY)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

1,482 substations (24.8%) High/Critical: **1121** (75.6%) Avg R: **0.553** Avg E: **0.687** / 0.10



🌱 Light / Agricultural / Rural

Est. VoLL: ¥100–400/kWh (JPY)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

1,495 substations (25.0%) High/Critical: **944** (63.1%) Avg R: **0.530** Avg E: **0.663** / 0.10

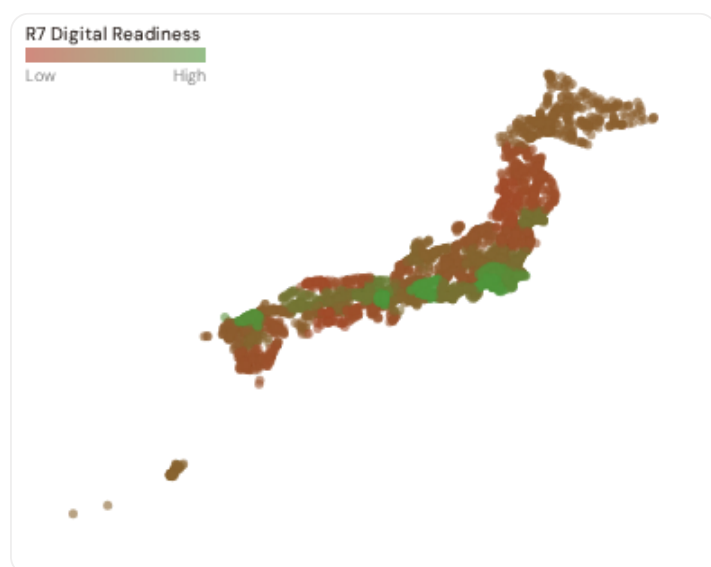


Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **1467 substations** combine capital-intensive economic fabric ($R3 \geq 1.15$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **TEPCO (582), Hokkaido (403), Kansai (230)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 8.0%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Prefecture Digital Readiness Ranking

Prefectures ranked by average R7 modifier — lower values indicate weaker digital infrastructure (MIC digital infrastructure sub-dimensions). Cross-referenced with average R_{median} to identify prefectures combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Akita Δ high R		0.9900
2	Kochi Δ high R		0.9900
3	Shimane		0.9900
4	Yamagata		0.9904
5	Iwate		0.9910
6	Tokushima Δ high R		0.9910
7	Tottori		0.9916
8	Miyazaki Δ high R		0.9916
9	Aomori		0.9916
10	Fukushima Δ high R		0.9922
11	Kagoshima Δ high R		0.9922
12	Wakayama Δ high R		0.9922
13	Fukui		0.9928
14	Oita		0.9928
15	Niigata		0.9934

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Akita** has the lowest average R7 (0.9900), while **Tokyo** leads at 1.0218 — a spread of 0.0318 across the R7 range. **11 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

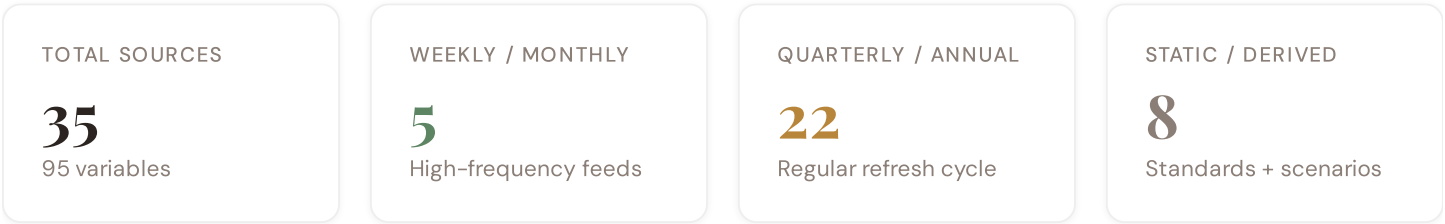
B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0012, median = 0.9982; 0 substations classified HIGH cyber-exposure; 2236 blind-spot substations (High/Critical risk + below-median R7); 89 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **35 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

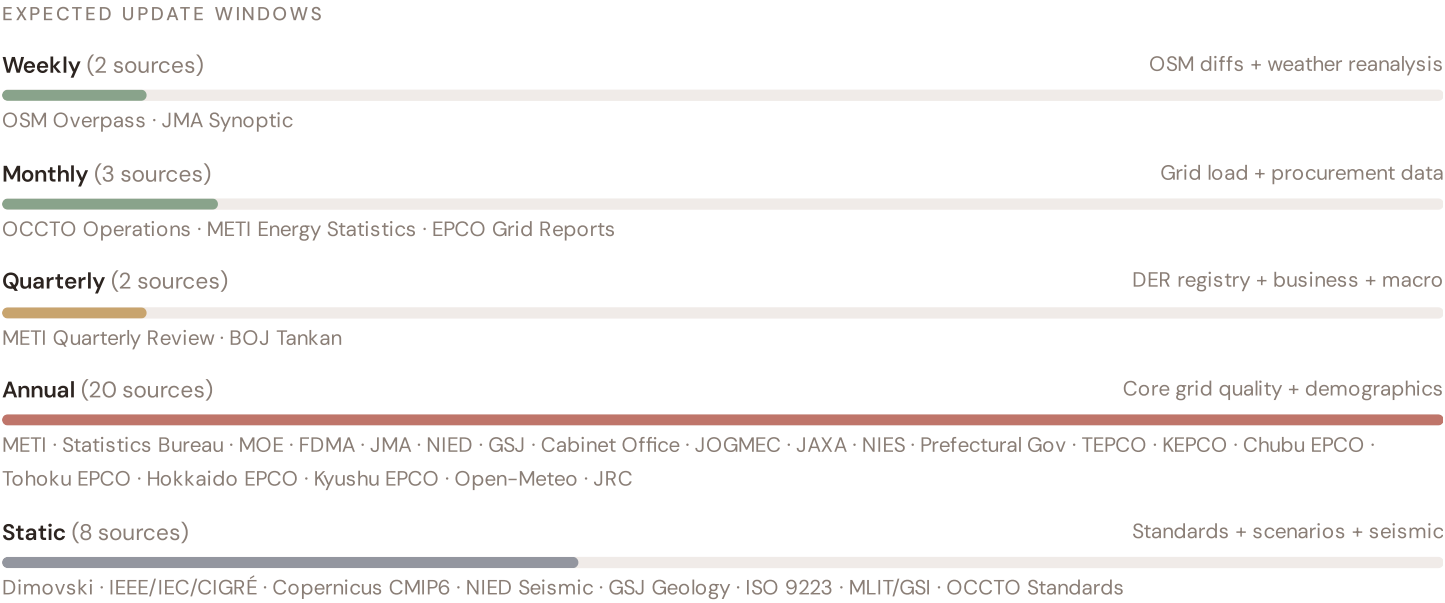


Data Source Registry — June 2026

JEPX	JEPX Wholesale Electricity Market Data	DAILY	Area	3 vars
OCCTO	OCCTO Area Supply-Demand Statistics	MONTHLY	EPCO Territory	3 vars
OpenStreetMap	OpenStreetMap Overpass API — Japan Grid	CONTINUOUS	Site-level	4 vars
JAXA	JAXA Satellite-Derived Climate Data	QUARTERLY	Regional	3 vars
BOJ	BOJ Regional Economic Reports (Sakura Report)	QUARTERLY	Regional	3 vars
ISEP	ISEP Energy Statistics	ANNUAL	Prefecture	2 vars
METI	METI Electricity Business Statistics	ANNUAL	EPCO Territory	3 vars
SBJ	SBJ Population Census / Current Population Survey	ANNUAL	Prefecture	4 vars
TEPCO	TEPCO Grid Reliability Reports	ANNUAL	EPCO Territory	4 vars
Kansai	Kansai Electric Reliability Data	ANNUAL	EPCO Territory	3 vars
Chubu	Chubu Electric Power Reliability Reports	ANNUAL	EPCO Territory	3 vars
Kyushu	Kyushu Electric Power Reliability Data	ANNUAL	EPCO Territory	2 vars
Hokkaido	Hokkaido Electric Power Reports	ANNUAL	EPCO Territory	2 vars
Tohoku	Tohoku Electric Power Data	ANNUAL	EPCO Territory	2 vars
Hokuriku	Hokuriku / Shikoku / Chugoku / Okinawa EPCO Reports	ANNUAL	EPCO Territory	2 vars
NISC	NISC Cybersecurity Reports	ANNUAL	National	2 vars
MHLW	MHLW Regional Health Statistics	ANNUAL	Prefecture	3 vars
MIC	MIC Digital Transformation Index	ANNUAL	Prefecture	3 vars
Cabinet	Cabinet Office GDP / Regional Economic Accounts	ANNUAL	Prefecture	3 vars
FDMA	FDMA Fire and Disaster Management Statistics	ANNUAL	Prefecture	3 vars
METI	METI Smart Grid Demonstration Data	ANNUAL	EPCO Territory	3 vars
MEXT	MEXT University/Research Infrastructure Atlas	ANNUAL	Prefecture	2 vars
MIC	MIC Migration Statistics (Jumin-Kihon-Daicho)	ANNUAL	Prefecture	2 vars
NPA	NPA Crime Statistics and Security Reports	ANNUAL	Prefecture	1 var

METI	METI Equipment Age Survey / TEPCO Asset Reports	ANNUAL	EPCO Territory	3 vars
GSI	GSI Elevation and Terrain Data	STATIC	10m mesh	3 vars
ISO	ISO 9223 / JIS Z 2381 Atmospheric Corrosion Maps	STATIC	Prefecture	1 var
AIST	AIST Geological Survey Data	STATIC	Regional	2 vars
IEEE	IEEE C57.91 Thermal Loading Standards	STATIC	Equipment-level	2 vars
IEC	IEC 61850 / JEC Standards Compliance	STATIC	Equipment-level	2 vars
NIED	NIED J-SHIS Seismic Hazard Maps	UPDATED PERIODICALLY	250m mesh	2 vars
JMA	JMA Seismological Bulletin / Typhoon Statistics	REAL-TIME + ANNUAL	National/Regional	3 vars
MLIT	MLIT Flood and Landslide Hazard Maps	UPDATED PERIODICALLY	Municipality	3 vars
NRA	NRA Nuclear Safety Reports	AS NEEDED	Site-level	3 vars
MIC	MIC Housing and Land Survey	QUINQUENNIAL	Prefecture	3 vars

C.1 Update Frequency Timeline



C.2 Fleet Score Snapshot

FLEET SIZE

5,981

substations scored

MEDIAN R

0.570

P5=0.460 · P95=0.731

HIGH + CRITICAL

4955

82.8% of fleet

HIGH CONFIDENCE

86%

5144 subs · CI/R < 0.50



No primary data sources have been updated since the v4.0.2 launch baseline. All **5,981 substation scores remain unchanged** this edition. The fleet median R of 0.570 (mean 0.578) reflects a distribution skewed toward the lower bands, with 0.0% in Low and 17.2% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across PO, P1, and P2 releases

F1	NEW	New T component aPABCCCH Energy Transition Exposure (T1)	§2, §4
F2	NEW	New R6a aPABCCCH Restoration Speed Modifier (CAIDI-based)	§5
F3	ENHANCED	R3 enhanced aPABCCCH Energy Poverty Vulnerability (V_socio)	§5
F4	ENHANCED	R4 enhanced aPABCCCH Graph-theoretic betweenness + bridge detection	§5
F5	ENHANCED	R2 enhanced aPABCCCH Climate Trajectory (CMIP6 SSP2-4.5)	§5
F6	ENHANCED	R7 Digital Readiness — Prefecture-level MIC digital infrastructure model	§5
L1	NEW	New I5, I7—I9 metrics aPABCCCH thermal, load, corrosion, hydrogeo	§2, §8
L2	ENHANCED	E2 Innovation Enrichment — MEXT R&D + university density	§6
L3	ENHANCED	V_socio Fiscal Enrichment — MIC + MOF + energy price indices	§5
L4	ENHANCED	R3 Demographic Shift Amplifier — SBJ net migration	§5
L5	DATA	95/95 variables operational (100%). 35 data sources total.	§8, §12
G1	DATA	d02 ISEP upgraded to quarterly registry — Prefecture-level DER registry	§12
G2	DATA	d05 AIST/GSJ upgraded to live API — Prefecture-level geological data	§12
G3	DATA	d06 OSM upgraded to Overpass API — 5,981 real substations	§12
G4	NEW	R6b Network Topology modifier aPABCCCH centrality + ring analysis from OSM graph	§5
G5	NEW	Network & Topology layer (H): 7 variables — OCCTO Grid Development Plan, ring analysis	§12
G6	NEW	Output Scores layer (I): 7 variables aPABCCCH risk_score, ETTC, stationary probs	§12

SECTION D — MONTHLY DEEP-DIVE

The Osaka–Aichi Corridor: Where Hydroelectric Dominance Meets Grid Modernization

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Osaka–Aichi corridor · **002** DER saturation hotspots (Tokyo/Fukuoka) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan–Rural economic impact disparity · **005** Nuclear fleet dependency and transition risk · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Prefecture Grid Health Cards · **010** Academic validation and peer feedback · **011** Asia–Pacific replication feasibility · **012** Year in review.

D.1 Why the Osaka–Aichi, Why Now

OSAKA–AICHI
SUBSTATIONS

403

43 EHV · 360 HV

HIGH BAND

334 + 69

100.0% of corridor

CORRIDOR MEDIAN R

0.649

Fleet median: 0.570

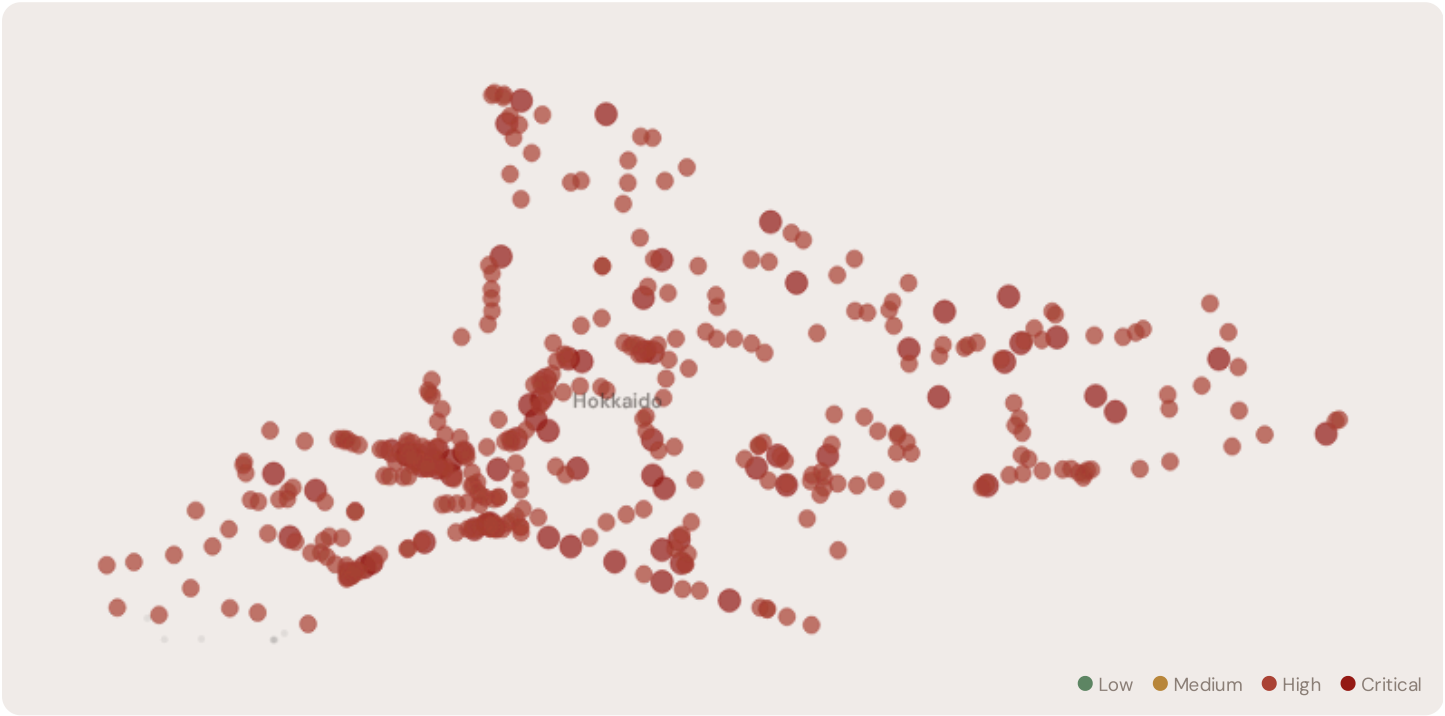
WORST PREFECTURE

Hokkaido

Avg R: 0.662 · 403 substations

The Hokkaido hosts **403 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **334 substations (82.9%)** are classified High and **69** Critical, with only **0** in the Low band. 92% of corridor substations score above the national fleet median of 0.570. The corridor median R of 0.649 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PREFECTURE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
JP_HOKKAIDO_03428	Hokkaido	0.884	Critical	0.485	0.810	0.359	0.783	0.559	0.362	V,E
富川変電所	Hokkaido	0.880	Critical	0.485	0.810	0.266	0.783	0.640	0.362	V,E
Saroma Substation	Hokkaido	0.877	Critical	0.485	0.810	0.336	0.783	0.625	0.362	V,E

SUBSTATION	PREFECTURE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
明野変電所	Hokkaido	0.862	Critical	0.485	0.810	0.262	0.783	0.634	0.362	V, E
豊沼変電所	Hokkaido	0.838	Critical	0.485	0.810	0.335	0.783	0.409	0.362	V, E
JP_HOKKAIDO_01650	Hokkaido	0.833	Critical	0.485	0.810	0.300	0.783	0.650	0.362	V, E
Shiroishi Substation	Hokkaido	0.823	Critical	0.485	0.810	0.293	0.783	0.625	0.362	V, E
Kanbetsu Substation	Hokkaido	0.821	Critical	0.485	0.810	0.376	0.783	0.368	0.362	V, E
東静内変電所	Hokkaido	0.818	Critical	0.485	0.810	0.368	0.783	0.648	0.362	V, E
JP_HOKKAIDO_01260	Hokkaido	0.813	Critical	0.485	0.810	0.341	0.783	0.613	0.362	V, E
... 393 additional substations — explore full corridor in Digital Twin →										

D.3 Component Analysis

COMPONENT AVERAGES — HOKKAIDO VS FLEET



MODIFIER AVERAGES — HOKKAIDO VS FLEET

R3 Consequence	1.184	fleet 1.109 ↑
R4 Graph Crit.	1.057	fleet 1.054 ↑
R6a Restoration	1.051	fleet 0.993 ↑
R6b Seismic	1.008	fleet 1.060 →
R7 Digital Read.	0.996	fleet 1.001 →

The dominant risk driver across the Hokkaido corridor is **V (Voltage)**, averaging 0.810 against a fleet average of 0.743 — occupying 810% of its weight ceiling (w = 0.10). The second contributor is **E (Economic)** at 0.783 vs fleet 0.630 (783% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.184 (fleet: 1.109), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.008, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Prefecture Breakdown

The region regions-level breakdown reveals significant intra-regional variation. **Hokkaido** leads with a mean R of 0.662 across 403 substations, while **Hokkaido** scores 0.662 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Hokkaido is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

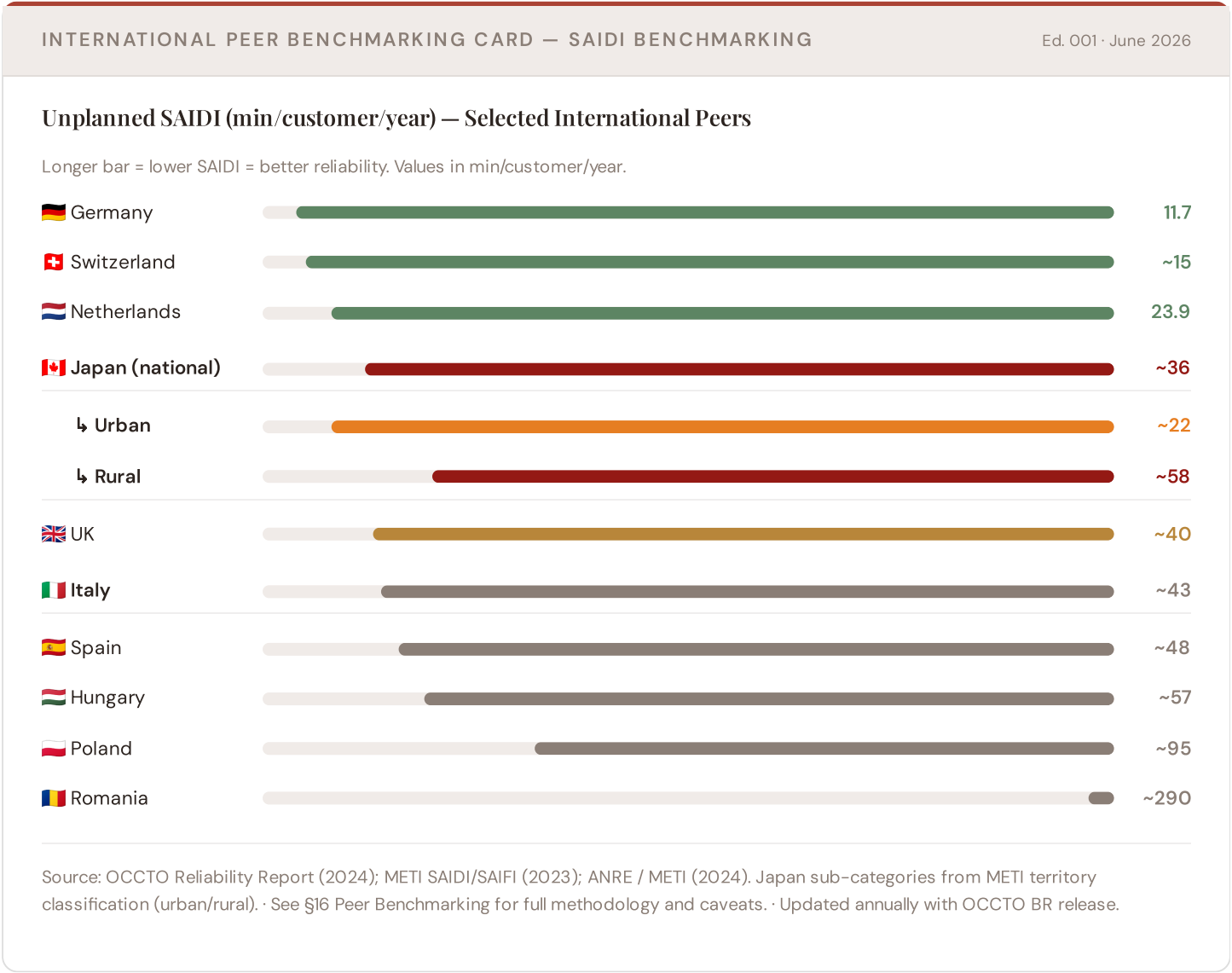
199 Hokkaido substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Hokkaido sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not

only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

International Peer Benchmarking Card

How this works: Each edition includes one International Peer Benchmarking Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Osaka–Aichi corridor analysis hinges on Japan's continuity performance relative to international peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Asia-Pacific (003), CMIP6 climate hazard vs East Asian peers (004), etc. The underlying data updates annually with the OCCTO Benchmarking Report release.



This month's key insight: The Hokkaido corridor deep-dive shows **403 substations** (of 403) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.485 is **1.3× the fleet average** of 0.371. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 10 EPCO Regions

Where does the Osaka–Aichi sit within the national landscape? Regions sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

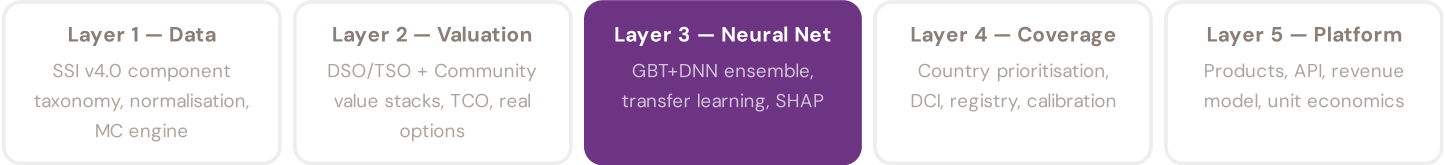


The Hokkaido ranks **#1 of 10 region regions** by mean R_{median}, placing it among the upper tier of national risk. The three most stressed are Hokkaido, Okinawa, Chubu, while the three most resilient are Hokuriku, Chugoku, Tohoku. 5 of 10 score above the fleet average of 0.578. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the TEPCO/OCCTO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



LAYER 3 §0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall:

starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

- SHAP: game-theoretic feature attribution per prediction
- Waterfall: fleet average → substation prediction
- Direct traceability to SSI components and data sources
- Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 5,981 substations with complete feature vectors. Average CI_width of 0.201 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 0 Low, 1026 Medium, 4753 High, 202 Critical — ensures balanced training across all risk bands.

Coming next month: **LAYER 3** Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

Q3 2026 — 2 Quarterly Sources

METI Quarterly Review, BOJ Tankan are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **SBJ Population Census / Current Population Survey** (4 variables → Population, elderly ratio, density, household counts by prefecture).

FLEET SPOTLIGHT

Hokkaido: Highest Risk Concentration

Hokkaido has the highest concentration of High/Critical substations: **403 of 403 (100%)** are in the upper risk bands — avg R = 0.662. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Japanese utility, METI, OCCTO, TEPCO, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 003

Published 11 June 2026 · SSI Index v4.0.2 · 5,981 substations · ~11,600 grid lines · 6 components · 20 metrics · 5 modifiers
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5